

What are the various systems and subsystems required to deliver this functionality?

- Systems and Subsystems Required
 - User-Facing Systems
 - Core Backend Systems
 - Banking & External Integration
 - Risk, Compliance & Operations
 - Infrastructure & Platform Services

What kind of information is exchanged between systems? Consider frequency, speed, volume, and consistency requirements.

- Key Data Types
 - User identity & KYC data
 - Bank account tokens (never raw account numbers)
 - Transaction requests & responses
 - Fraud risk scores
 - Settlement instructions
 - Audit & compliance logs

What are the major potential failure points in this system?

- **Technical failures**
- **Integration failures**
- **External dependency failures**

- Technical Failures
 - Payment processing node crash
 - Database latency or unavailability
 - Cache inconsistency
 - Network partition

Integration Failures

- Bank API downtime
- Settlement system lag
- Third-party API rate limits

External Dependency Failures

- Partial bank outages

- Telecom/network failures
- Regulatory reporting endpoint failures
- Sudden 10x traffic spikes

**What mitigation strategies would you design to prevent or reduce failures? For Eg :
Redundancy, Circuit breakers etc**

➤ Architecture-Level

- Active-active multi-region deployment
- Stateless microservices
- Horizontal auto-scaling
- Sharded & replicated databases

Resilience Patterns

- Circuit breakers for bank APIs
- Retry with exponential backoff
- Bulkheads for critical services
- Idempotent transaction handling

Data Protection

- Write-ahead logs
- Dual ledger recording
- Guaranteed message delivery

What contingency plans are required when failures do occur?

- **Manual reconciliation**
- **Deferred settlements**
- **User communication strategies**

➤ Financial Continuity

- Deferred settlements when banks are down
- Shadow ledger for pending transactions
- Manual reconciliation workflows

Operational Response

- Incident dashboards for regulators
- Automated alerts to ops teams

- Graceful service degradation

User Communication

- Real-time transaction status updates
- Clear “Pending – Bank Delay” messaging
- Push/email/SMS notifications

How should the system behave differently under:

- **Normal daily load**
- **Peak national events**

➤ Normal Daily Load

- Fully synchronous real-time processing
- Real-time fraud scoring
- Immediate settlements where possible

Peak National Events

- Aggressive auto-scaling
- Asynchronous settlement processing
- Throttling non-critical features (reports, analytics)
- Prioritization of payment traffic over admin traffic

What architectural trade-offs would you make between:

- **Consistency vs availability**
- **Cost vs reliability**
- **Real-time vs eventual processing**
- **Security Depth vs User Experience**

➤ Consistency vs Availability

- Availability prioritized for payment acceptance
- Eventual consistency allowed for settlements & reports

Cost vs Reliability

- Higher cost justified for:
 - Multi-region redundancy
 - 99.999% availability

- Commodity infra for analytics & reporting

Real-Time vs Eventual Processing

- Real-time: Payments, balance validation
- Eventual: Reconciliation, regulatory reports

Security Depth vs User Experience

- Strong backend security (tokenization, encryption)
- Minimal friction UX (biometric auth, device binding)

8. Refer to similar global implementations and highlight the USPs and additional features that your solution recommends. Highlight the impact and ROI of these additions.

- Unified Payments Interface (India)

Recommended Additional Features

Feature	Impact	ROI
Offline QR payments	Inclusion & resilience	High adoption
Smart fraud ML models	Reduced fraud loss	Direct savings
Cross-border instant rails	International expansion	New revenue streams
Open APIs for fintechs	Ecosystem growth	Innovation at scale
Merchant analytics	Better merchant retention	Higher transaction volumes

Final Summary

This architecture delivers:

1. National-scale real-time payments
2. Extreme resilience & fault tolerance
3. Regulatory-grade security
4. Future-ready international expansion